NATURAL RESOURCE CENSUS

Annual Assessment of Land Cover and Land Use using Multi-Temporal data

Technical Document on –

Land Cover and Land Use Database for Dissemination through Bhuvan

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Novemeber, 2025

Document Control Sheet

01.	**Security Classification**	Unrestricted			
02.	**Distribution**	General			
03.	**Document Version**	**(a) Issue no.** 01		**(b)Revision &Date**	--
04.	**Document Type**	Technical Document – Ver. 1			
05.	**Document Control Number**	**NRSC-RSA-SR&LUMG-SRLMD-NOV 2025-TR-0002929-V1.0**			
06.	**Title**	Annual Assessment of Land Cover and Land Use using Multi-Temporal data for Dissemination through Bhuvan			
07.	**Particulars of Collation**	**Pages - 16**	**Figures 3**	**Tables - 1**	**References - 6**
08.	**Author(s)**	Scientists and Research scientists from Soil Resources and Landuse Monitoring Division (SR&LMD)			
09.	**Affiliation of Authors**	SRLM Division, SR & LRUM Group, Remote Sensing Applications Area, NRSC			
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11.	**Originating Unit**	SRLMD, SR&LUMG, RSA A			
12.	**Sponsor(s)/ Name & address**	NNRMS, ISRO, Bengaluru			
13.	**Date of Initiation**	October, 2025			
14.	**Date of Publication**	October, 2025			
15.	**Abstract**	The geo-spatial dataset titled “Annual Assessment of Land Cover and Land Use using Multi-Temporal Data” is generated primarily using month-wise Resourcesat-2/2A ortho-rectified AWiFS data, complemented by Sentinel-1 and Sentinel-2 open-source datasets for a given year. The entire database is prepared by NRSC, ISRO under the Natural Census Programme, and this LCLU layer has been produced annually since 2005–06. The project aims to generate crop area information at the end of the Kharif, Rabi, and Zaid cropping seasons, along with an integrated annual Land Cover and Land Use (LCLU) dataset. The data are produced using a rule-based semi-automated approach, following an 18-class LCLU classification schema. The final outputs, provided in GeoTIFF format, are disseminated through Bhuvan – ISRO’s Geoportal.			

EXECUTIVE SUMMARY

Land Cover and Land Use (LCLU) data provide essential information on the Earth's surface features and how they are utilized by humans. These datasets support planning, resource management, and environmental monitoring by depicting spatial patterns of natural vegetation, water bodies, built-up areas, and agricultural land. Capitalizing on the potential of Remote Sensing data, the National Remote Sensing Centre (NRSC), undertakes a comprehensive national-level annual assessment of Land Cover and Land Use as part of its contributions to the National Natural Resource Management System since year 2005-06 onwards. LCLU is designated as one of the 14 National Fundamental Geospatial Data Themes under National Geospatial Policy, making its regular updating and integration central to India's spatial data infrastructure. The NGP – 2022 policy mandates that nodal agencies (like NRSC/DOS and MoEF&CC) regularly assess and publicly disseminate LCLU data (e.g. via the Bhuvan geo-portal) to support planning & environmental governance.

The primary objectives of this endeavor encompass the following:

- To rapidly assess the National-Level Annual LCLU using multi-temporal and multi-sensor satellite data.
- To compile crop sown area data for the Kharif and Rabi seasons after each season ends.
- To generate derivative products required for the National Information System for Climate & Environment Studies (NICES) programme.

To achieve above objectives, a rigorous land use classification framework encompassing 18 distinct classes and semiautomatic method of mapping the National LCLU at 56m resolution (commensurate to 1: 250,000 scale) is developed. Since 2005, the National Remote Sensing Centre (NRSC) has been undertaking annual assessment of LCLU for the country under Natural Resources Census (NRC) Programme of ISRO with funding under National Natural Resources Management System (NNRMS). Towards this multi-temporal and multi-sensor satellite data, primarily from AWiFS and supplemented by Open Source data (e.g. Sentinel – 1 & 2, Landsat etc.) have been analysed to create LCLU layer for the entire country. This involves pre-processing of data, followed by classification through the implementation of semi-automated rule based techniques. The outcome of this project encompasses preparation of geo-spatial thematic dataset for the entire country on 56m resolution comprising of:

- Annual Land cover and land use data,
- Annual Kharif and Rabi crop sown data,
- Monthly crop sown area for January, August, September and December
- Reprocessed LCLU inputs (5km*5km grid) requisite for modelling purposes e.g., NICES programme and meteorological forecasting models.

Commencing its journey in 2005–2006, the analysis reveals a steady rise in cropland and a sharp decline in fallow land. Kharif and Rabi cropland expanded while double/triple cropping rose significantly. Built-up area grew modestly, and water bodies increased too. An atlas (https://www.nrsc.gov.in/nrscnew/assets/pdf/atlas/LCLU/LCLU%20Atlas%20Final%20With%20Cover_March2024.pdf) is published after compilation of LCLU from 2005–2006 to 2022-2023. This provides an insights into India's land cover and land use dynamics and it can serve a role in promoting informed decision-making and fostering sustainable development practices across the country. Till date 200+ users have used these LCLU datasets for various research, modelling and developmental monitoring activities by Academic Institutions (e.g. IITs, IIITs, IISER, NITs, IIM. research organizations etc.) and government organization (ICAR Institutes, NIH, WII, ICFRE, GSI, MoLJ etc.).

1. INTRODUCTION

Effective decision-making in a modern, diverse country like India - characterized by varied landscapes, ecosystems, and land-use patterns - requires reliable information across multiple interrelated domains. Among these, Land Cover and Land Use (LCLU) form critical components for environmental management, developmental planning, and sustainable resource utilization.

Although the terms Land Cover and Land Use are often used interchangeably, their meanings differ. Land Use refers to the purpose for which land is utilized - such as agriculture, settlements, forestry, or mining. According to the Food and Agriculture Organization (FAO), land use encompasses the totality of arrangements, activities, and inputs that people apply to a particular type of land cover. In contrast, Land Cover denotes the observable physical and biological features on the Earth's surface, including vegetation, forests, grasslands, scrubs, water bodies, and barren areas. Accurate and up-to-date land use data have become increasingly essential as India addresses challenges such as unplanned urban expansion, degradation of environmental quality, loss of fertile agricultural lands, and the depletion of wetlands and wildlife habitats. Reliable LCLU information enables both governmental and private organizations to assess ongoing transformations and to develop informed strategies for sustainable development and environmental conservation.

The classification and mapping of LCLU using remote sensing data play a pivotal role in monitoring and managing changes in the environment. Satellite sensors capture data across multiple spectral bands, allowing for the identification of various land cover types through their unique spectral signatures. This capability supports the generation of accurate and consistent land cover maps that are vital for urban planning, natural resource management, and environmental monitoring. Furthermore, the temporal availability of remote sensing data enables change detection - facilitating assessments of deforestation, urban sprawl, or agricultural transitions over time - and thereby supports evaluation of land management policies and practices.

Recognizing the significance of LCLU, the National Geospatial Policy (NGP) – 2022 identifies it as one of the 14 National Fundamental Geospatial Data Themes, emphasizing its regular updating and integration into the national geospatial data infrastructure. As per NGP-2022, nodal agencies (NRSC/DoS along with MoEF&CC) are mandated to periodically assess and disseminate LCLU data through platforms like the Bhuvan Geoportal, thereby supporting evidence-based planning and environmental governance.

With these in background and alignment with this policy, a national-level LCLU mapping initiative has been undertaken under the National Natural Resources Management System

(NNRMS) programme of ISRO-DoS. The project aims to facilitate systematic spatial mapping and monitoring of India's land cover and land use, contributing to informed policy formulation and sustainable resource management.

2. OBJECTIVES

The Land Cover and Land Use (LCLU) geo-spatial layer for the country is generated annually to provide consistent, up-to-date information on land dynamics. The primary objectives of this initiative are as follows:

- Rapid assessment of the national-level annual LCLU using multi-temporal and multi-sensor satellite data.
- Compilation of crop sown area statistics for both Kharif and Rabi seasons upon completion of each agricultural cycle.
- Generation of derivative products to support the National Information System for Climate & Environment Studies (NICES) programme and related applications in environmental and resource management.

3. CLASSIFICATION SCHEMA

The annual assessment of Land Cover and Land Use (LCLU) has been undertaken since 2005–2006, employing a standardized 18-class classification system. This schema has been applied consistently across all years of analysis to ensure both temporal and thematic uniformity at the national scale. The classification framework encompasses a comprehensive range of land cover and land use categories, including various agricultural classes, forest classes, water bodies, wastelands, built-up areas, and other surface features. Each class is delineated based on its distinct spectral, spatial, and contextual characteristics derived from multi-temporal satellite imagery, enabling accurate identification and discrimination of diverse land features.

Maintaining the same classification schema throughout the study period ensures comparability of results, minimizes interpretation bias, and supports robust temporal change detection and trend analysis. This methodological consistency provides a strong foundation for assessing long-term landscape dynamics and understanding spatio-temporal transformations in India's land systems.

A summary of the adopted LCLU classification schema along with a brief description of the individual classes is presented in Table–1 below:

Table – 1: LCLU Classification scheme used for Annual LCLU project

Sl.	LCLU Class	Description
1	Built-up land	Residential areas, industries, airport and other impermeable surfaces generated by anthropogenic activity.
2	Kharif crop land	Seasonal Crop land with crops grown during June to November period of agricultural calendar year.
3	Rabi crop land	Seasonal Crop land with crops grown during November to April period of agricultural calendar year.
4	Zaid Crop land	Seasonal Crop land with crops grown during April to June period of agricultural calendar year.
5	Double/Triple/annual crop land	Land with crops grown in more than one season specified above. This will also include annual crops.
6	Current Fallow land	These are the lands, which are taken up for cultivation but are temporarily allowed to rest, un-cropped for one or more season, but not less than one year
7	Plantation / Orchard	These are the areas under tree crops that are artificially planted adopting agricultural management techniques.
8	Evergreen / Semi-evergreen Woodland	This category comprises of trees (>2m tall), which are predominantly remain green throughout the year. It includes both coniferous and tropical broadleaved evergreen species. Semi- evergreen is a woodland type that includes a combination of evergreen and deciduous species with the former dominating the canopy cover.
9	Deciduous Woodland	Woodland types that are predominantly composed of tree (>2m tall) species, which shed their leaves once a year. It may also include tree clad area with tree cover lying outside the notified forest boundary areas that are herbaceous with a woody appearance
10	Degraded Woodland	Land covered with tree species (more than 2m tall) which are Evergreen / Deciduous in nature with relatively decreased density of
11	Littoral/Swamp/Mangroves	Areas with seasonal or permanent water ponding (with or without vegetation) excluding the water bodies. These include ox-bow lakes, tidal flat/mud flat, mangrove, salt marsh/marsh vegetation and other hydrophytic vegetation
12	Grassland Land	Areas with seasonal or perennial grasses occur naturally.
13	Shifting cultivation	These are the areas where woodlands are cleared and used for cultivation
14	Wastelands	These are barren lands with nil or little vegetation cover and includes areas like rocky areas, scrub lands, mining dumps, gullied lands, sand dunes etc.
15	Rann	These are the areas with very high concentrations of salts usually sourced from sea and occur near the sea coasts.
16	Water Bodies – maximum spread	This represents the maximum water spread in a water body like lakes, tanks, reservoirs, rivers, etc.
17	Water Bodies – minimum spread	This represents the least water spread in a water body like lakes, tanks, reservoirs, rivers, etc.
18	Snow covered / Glacial areas	Land under snow cover / ice, mostly permanent.

4. DATA USED

The satellite remote sensing data used for annual Land Cover and Land Use (LCLU) assessment were primarily derived from the AWiFS sensor and open-source datasets, acquired on a monthly basis to ensure the selection of the best possible cloud-free imagery. For national coverage, approximately 21 full AWiFS scenes or equivalent imagery from open source platform are required. However, due to frequent cloud cover, particularly during the Kharif season (August–September), additional datasets were procured to minimize data gaps. In regions affected by persistent cloud conditions, Sentinel-1 and Sentinel-2 datasets were integrated to maintain temporal and spatial continuity. Additionally, RISAT-1 CRS and/or MRS data were also employed during the Kharif season to enhance.

Along with satellite observations, rainfall patterns and the geographical distribution of crops across meteorological subdivisions were analyzed to support accurate seasonal classification. The major crop seasons i.e. Kharif and Rabi were represented using multi-temporal satellite datasets covering the June–May agricultural calendar year. To address any residual data gaps in persistently cloudy regions, additional multispectral and microwave datasets from Landsat, Sentinel-1, and Sentinel-2A/B were incorporated.

Topographic parameters were derived from Carto DEM, SRTM DEM and other relevant elevation datasets to improve terrain-based analysis. Since satellite data alone may not fully distinguish spectrally similar land cover types, several ancillary datasets were used, including:

- National-level forest and wasteland spatial databases,
- 1:50,000 scale LCLU maps as reference layers, and
- Ortho-corrected high-resolution imagery available on Bhuvan, resampled to 56 m resolution to ensure geometric consistency.

Finally, state-level maps and zonal statistics were generated using geo-referenced state boundaries and delineated meteorological zones, enabling regional-level assessment and aggregation of LCLU statistics.

5. METHODOLOGY

5.1 PRE-PROCESSING

The satellite imagery used in the study was initially available in raw digital format, without any geometric or radiometric corrections. Therefore, a series of radiometric and geometric pre-processing steps were performed to ensure data consistency and analytical accuracy.
Radiometric correction involved two key steps:

- Sensor calibration, which converted digital numbers (DNs) into radiance values; and

- Reflectance estimation, which transformed radiance into top-of-atmosphere (TOA) reflectance values.

The various parameters needed for estimating spectral reflectance are: maximum and minimum radiances in the satellite sensor bands, mean solar exo- atmospheric irradiances, image acquisition time, solar declination, solar angles (zenith and azimuth) and mean Earth–Sun distance. Ortho-rectification and TOA correction were applied to all multi-temporal optical datasets. The corresponding sensor-specific calibration constants were retrieved and applied to generate TOA reflectance for each satellite scene.

Ortho-rectified satellite datasets were used on a monthly basis for analysis. Geometric correction was performed using the automated geo-rectification workflow in IMGEOS (NRSC). The registration accuracy was maintained within <1 pixel for plains and up to 3 pixels in hilly regions. Owing to the improved geometric precision achieved through the IMGEOS data processing chain, full-scene satellite images were adopted for subsequent analyses.

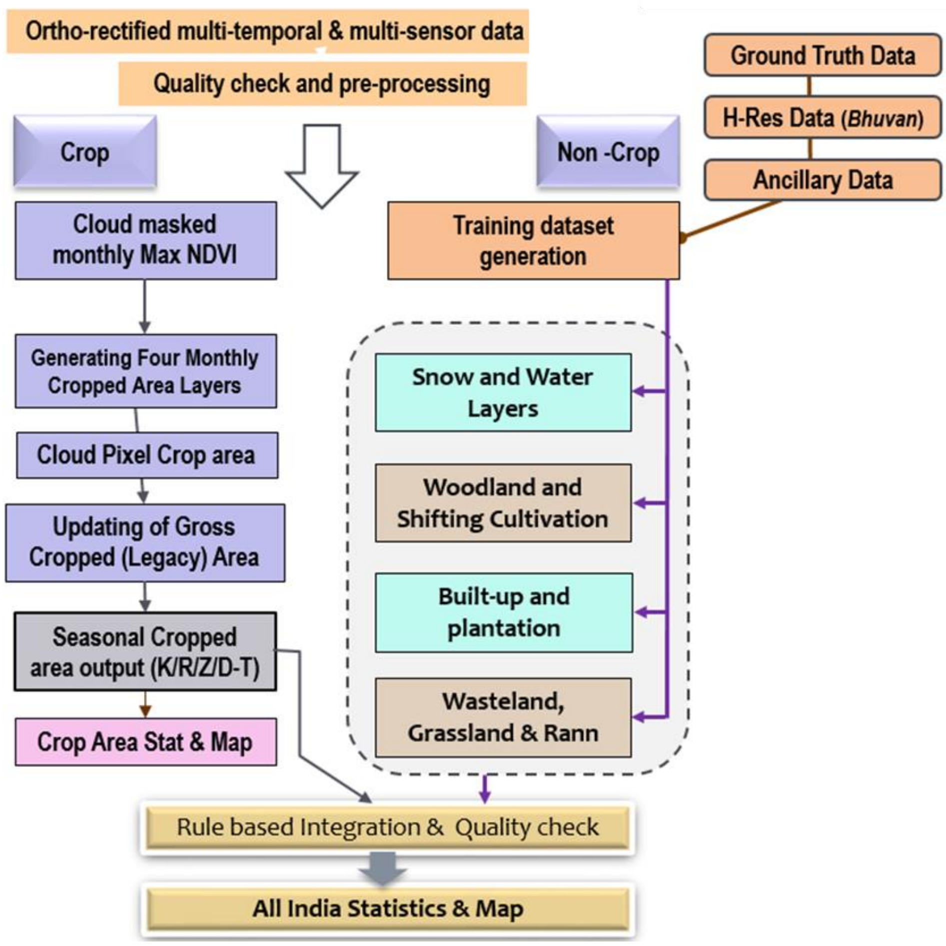

Figure 1: Overall Methodology Flowchart

After acquiring geometrically rectified and quality-assured datasets, TOA reflectance data were generated and used. The multi-temporal and multi-sensor imagery was then subjected to detailed pre-processing workflows. The processed datasets were subsequently classified using a suite of semi-automated, rule-based classification techniques, ensuring high thematic accuracy and spatial coherence across seasons and regions. The overall methodology followed in demonstrated in the below Figure-1

5.2 CLASSIFICATION OF SATELLITE DATA

The Land Cover and Land Use (LCLU) mapping methodology followed a multi-stage workflow integrating remote sensing and Geographic Information System (GIS) techniques, complemented by the analysis of climate parameters - particularly for agriculture-related land cover classes.

Initially, satellite imagery was acquired and pre-processed to ensure optimal data quality and geometric consistency. Monthly Normalized Difference Vegetation Index (NDVI) composites were generated from multi-spectral datasets, incorporating necessary radiometric corrections using a semi-automated processing chain. Each layer was extracted individually using a combination of multi-temporal and multi-sensor datasets, followed by post-processing to minimize noise and enhance spatial and spectral integrity. The pre-processed imagery was then subjected to classification following the methodology outlined in Figure–1. Specific classes such as Built-up, Snow/Ice, and Water Bodies were extracted using digital classification techniques, while Wasteland categories were delineated through interactive validation and contextual refinement based on neighboring land cover characteristics.

Legacy LCLU classes - such as evergreen woodland, deciduous woodland, degraded woodland, grassland, wasteland, and rann - were further refined by incorporating current-year information on crop patterns, built-up expansion, snow cover, shifting cultivation, and mangrove distribution. This iterative improvement process ensured that updated datasets accurately captured the evolving boundaries of LCLU classes. Ground truth data collected through field observations were utilized in defining training sets and validating classification outputs. For agricultural land, a threshold-based approach was employed to delineate crop-sown areas for the relevant months. Monthly classified outputs were integrated at the end of each crop season to generate comprehensive maps representing cropped area, current fallow, water, and snow cover for that period. The classified outputs from Kharif, Rabi, and Zaid seasons, along with dynamic land cover classes such as Built-up, Water Bodies, and Snow, were subsequently integrated using logical hierarchical rule sets to produce the final integrated LCLU product for the annual cycle.

All input and output datasets were standardized to a common spatial reference, ensuring uniform pixel size and consistent alignment. The standardized layers were then integrated using a rule-based GIS workflow, enabling accurate and coherent representation of land cover and land use dynamics across spatial and temporal scales.

6. Quality Assurance and Accuracy

To ensure consistency, reliability, and high accuracy in the derived Land Cover and Land Use (LCLU) products, a comprehensive Quality Assurance (QA) and Quality Control (QC) framework was implemented for both input and output datasets.

Quality Assurance (QA) of input data involved systematic verification of all satellite imagery, ancillary datasets, and derived indices to confirm geometric accuracy, radiometric integrity, and temporal consistency. The QA process relied on multiple reference sources, including ground truth data, high-resolution multi-spectral (MX) imagery, and legacy reference maps.

Similarly, Quality Control (QC) checks were performed on all intermediate and final output layers to ensure thematic accuracy, logical consistency, and alignment with reference datasets. Only QC-cleared outputs were approved for dissemination through the Bhuvan web portal, ensuring uniform quality standards at the national level.

Accuracy Assessment

The accuracy assessment of the final LCLU maps was performed using a stratified random sampling approach, ensuring adequate representation of all land cover and land use classes across the country. A set of random validation points was generated and evenly distributed among all LCLU categories, with a minimum of 25 points per class to maintain statistical robustness. These validation points were compared against mid- to high-resolution open-source datasets and verified using ground truth observations where available. The reference datasets served as the benchmark for evaluating classification accuracy, enabling the calculation of standard accuracy metrics.

This multi-tiered quality assurance and accuracy assessment framework ensured that the generated LCLU layers are both scientifically reliable and operationally consistent, supporting their use in national-level planning, environmental monitoring, and policy formulation.

7. RESULTS

The Integrated Annual Land Cover and Land Use (LCLU) Map of India has been prepared, depicting the spatial extent of various land cover categories for each year since 2005–06. The outputs undergo verification for class harmony, internal consistency, and temporal coherence with respect to the previous year’s data before the generation of final statistics. The finalized outputs are disseminated and visualized through the Bhuvan geoportal, providing maps and statistics at both national and state levels.

A sample LCLU Annual Map (2023–24) along with its statistical summary is presented in Figure 2. Similarly, Kharif and Rabi cropped area layers have been derived through digital analysis for all annual cycles since 2005–06, with examples for 2023–24 shown in Figure 3.

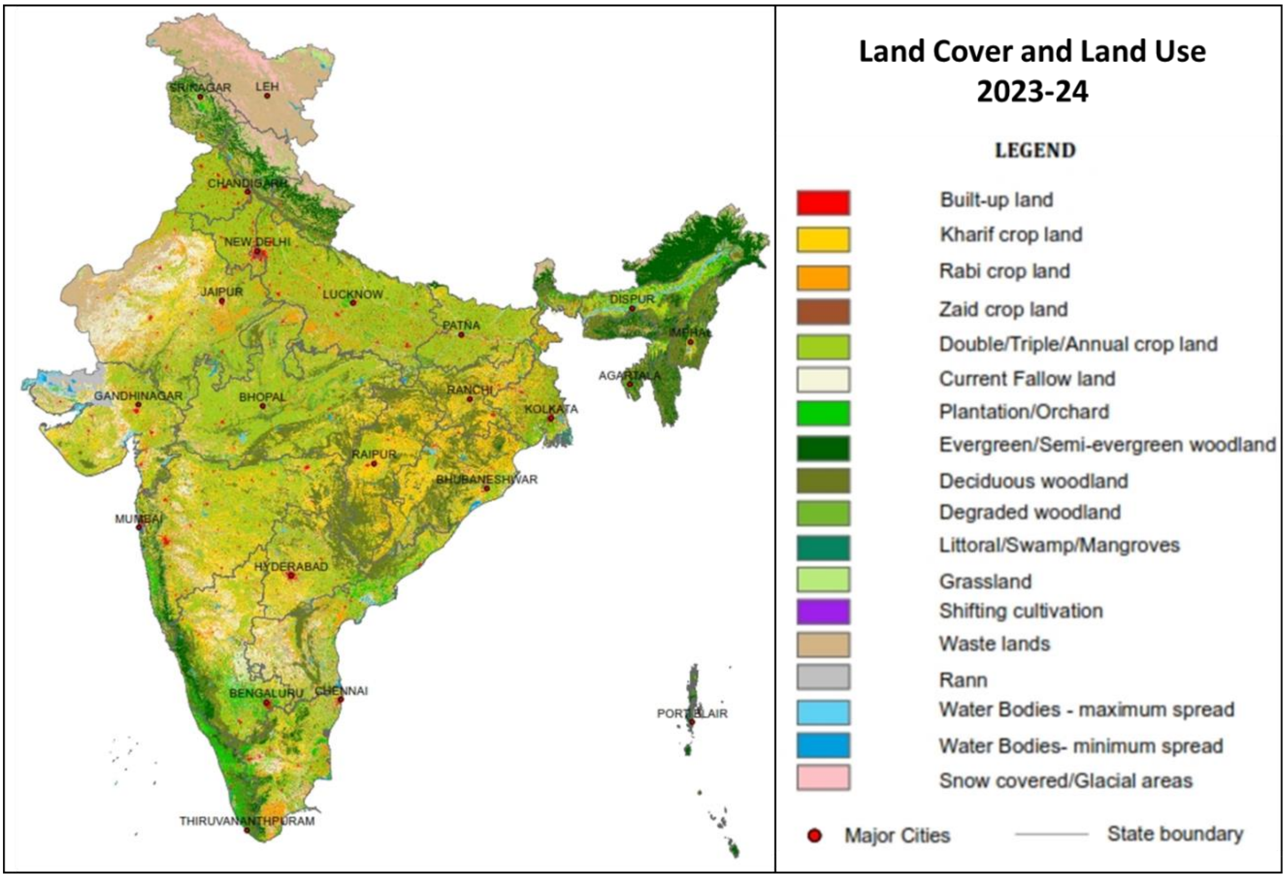

Figure 2: Annual Land Cover and Land Use map (2023-24)

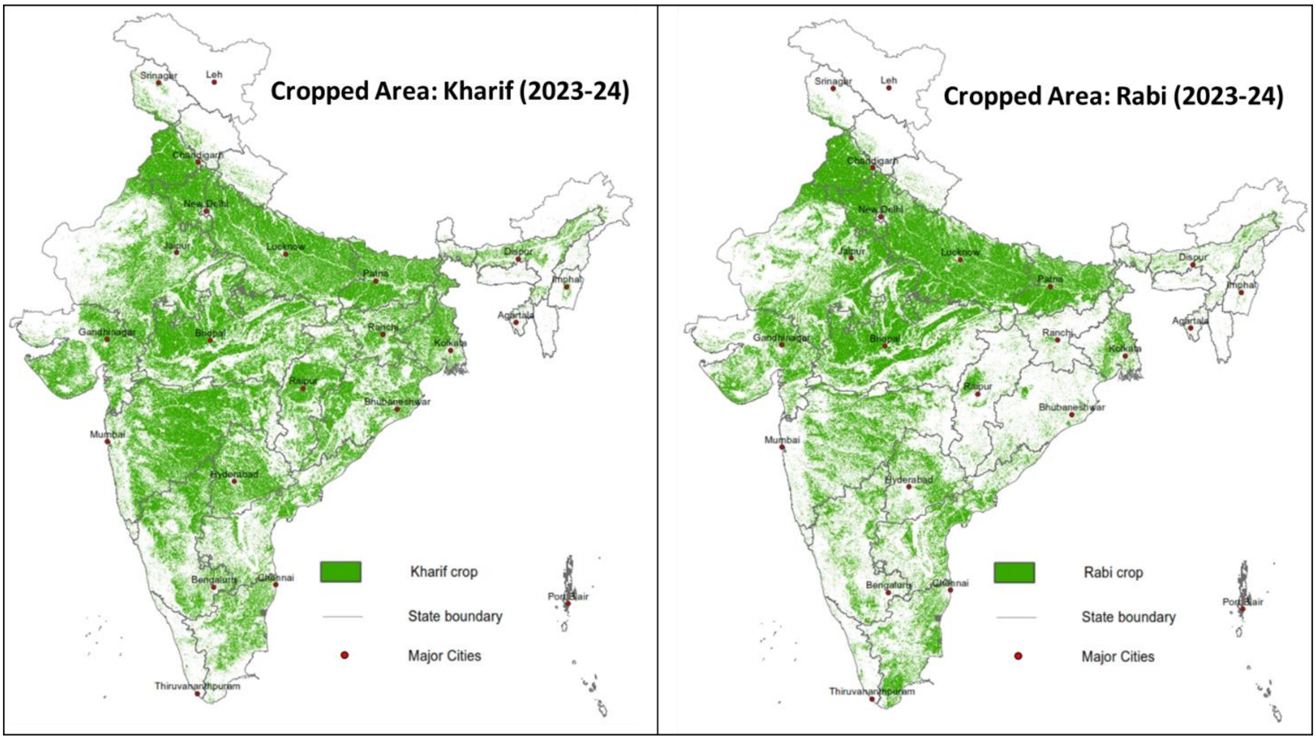

Figure 3: Estimated Kharif and Rabi cropped map (2023-24)

8. SUGGESTED USE

The LCLU maps are intended for broad-level applications across scientific, planning, and management domains, including:

- Scientific research on the carbon cycle, hydrologic cycle, energy budget studies, and weather or climate prediction.
- Siting of industries, Special Economic Zones (SEZs), and other infrastructure projects.
- Land improvement programmes and soil conservation initiatives.
- Watershed management and integrated water resources planning.
- Coastal zone management and related environmental monitoring.
- Enhancing agricultural productivity through crop planning and land use optimization.
- Formulation of development programmes by ministries and departments such as Agriculture, Environment & Forests, Water Resources, and others.
- Monitoring dynamic land cover classes, including surface water, forests, and wastelands.
- Climate impact assessment and related studies using mid-resolution annual LCLU datasets.

LIMITATIONS

- Similar spectral responses from different surface features, or dissimilar responses from similar features, can cause spectral confusion and lead to misinterpretation during analysis.
- While most LCLU classes can be identified with limited ground truth, certain classes - particularly in areas with mixed land use patterns - require more systematic ground verification due to the limited spatial resolution of satellite data.
- Identifying and classifying multiple uses occurring on the same land parcel is challenging. Consequently, generalization and grouping of multiple land use classes is often necessary.
- Some land cover categories may be present on the ground but not in sufficient areal extent to be detected from satellite imagery. Determining optimal detection parameters for specific applications is an iterative learning process.
- Due to variations in projection parameters, the area computed from interpreted GIS vectors may not exactly match ground-surveyed areas.
- Accurate analysis and interpretation of remote sensing data requires trained and skilled personnel.

DISCLAIMER

- The accuracy of different land cover and land use classes depends on the availability of suitable satellite imagery windows. The provision of such data is subject to currently operational and planned future ISRO satellites.
- The data and derived products are not intended for legal purposes and should not be used as such.
- Users are expected to exercise reasonable skill, care, and diligence when utilizing the information. NRSC/ISRO shall not be held liable for any loss, damage, or claims arising from the use of this information, and users shall indemnify NRSC/ISRO against such occurrences.

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CITATION

NRSC (2025), Land Cover and Land Use database on 1:250,000 scale, SRLMD, SR & LUMG, RSAA, National Remote Sensing Centre, ISRO, Hyderabad

ACKNOWLEDGEMENT

Use of data in any form is to be duly acknowledged as shown below:

I/we have used the Land Cover and Land Cover information on my/our research work from Natural Resources Census Project of National Remote Sensing Centre (NRSC), ISRO, Hyderabad, India

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